

Airline Data Analysis

Project Overview

This project focuses on analyzing airline data to extract business insights related to flight activity, revenue, customer behavior, and booking patterns.

The project consists of multiple dashboards built using Power BI, where each dashboard addresses a different analytical perspective and answers specific research questions.

Analysis Topic

The analysis focuses on airline operations, including:

- Flight distribution between destinations
- Revenue generation across routes
- Passenger distribution and movement
- Booking behavior over time

Research Questions

1. Which airports have the highest number of flights?
2. Which routes generate the highest revenue?
3. How are passengers distributed across destinations and airlines?
4. What are the booking patterns across months and days of the week?
5. Are there peak periods or trends in booking activity?

Analysis Explanation

Data Description:

The dataset used in this project contains information related to airline operations, including:

- Flights data (flight routes, departure and arrival airports)
- Bookings data (number of bookings and passengers)
- Airports data (airport names, cities, and countries)
- Airlines data (airline names and details)
- Ticket prices and revenue-related fields

This data enables analysis from multiple perspectives, including operational activity, financial performance, customer distribution, and booking behavior.

Data Cleaning and Preparations:

- Promoted headers to properly structure the dataset
- Renamed columns to ensure clarity and consistency
- Created calculated columns to support analysis
- The Data was already clean from nulls and inconsistencies

Data Model:

A relational data model was built to connect the different tables. The model includes:

- A relationship between bookings and flights
- Connections between flights and airports
- A link between airlines and bookings

Measures:

- Total Bookings
- Total Flights
- Total Passengers
- Total Revenue
- Average Ticket Price
- Revenue per Flight

Visualizations:

- Map to display geographic distribution of passengers

200

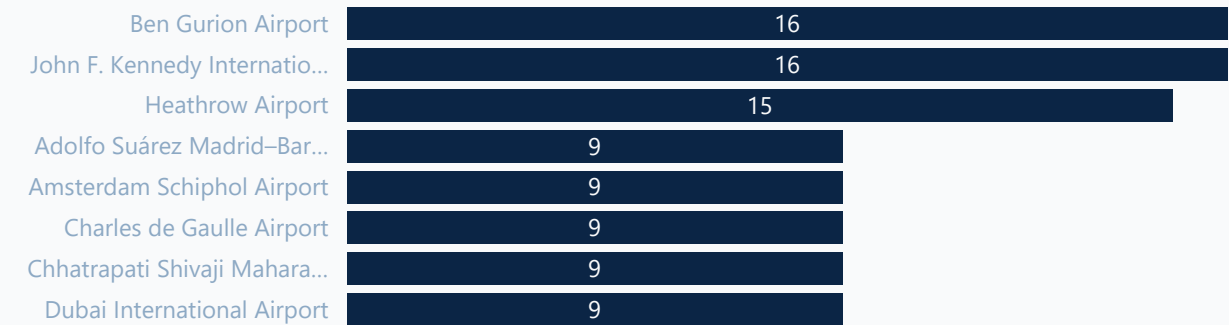
Total Flights

Flight Traffic Analysis

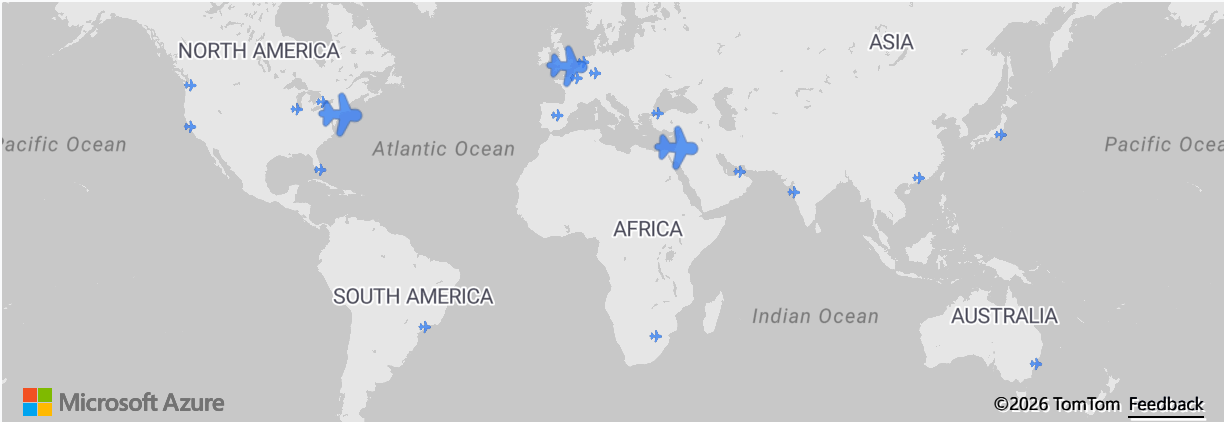
Arrival Airport

Departure Airport

Departure Airport



Departure Airport



Departure Airport	Arrival Airport	Total Flights
Ben Gurion Airport	John F. Kennedy International Airport	5
John F. Kennedy International Airport	Heathrow Airport	5
Ben Gurion Airport	Heathrow Airport	4
Heathrow Airport	Ben Gurion Airport	3
Heathrow Airport	John F. Kennedy International Airport	3
John F. Kennedy International Airport	Ben Gurion Airport	3
Adolfo Suárez Madrid-Barajas Airport	Ben Gurion Airport	2
Adolfo Suárez Madrid-Barajas Airport	Seattle-Tacoma International Airport	2
Amsterdam Schiphol Airport	Chhatrapati Shivaji Maharaj International Airport	2
Amsterdam Schiphol Airport	Seattle-Tacoma International Airport	2
Ben Gurion Airport	Tokyo Haneda Airport	2
Charles de Gaulle Airoort	O. R. Tambo International Airport	2

98K

Total Revenue

\$546.67

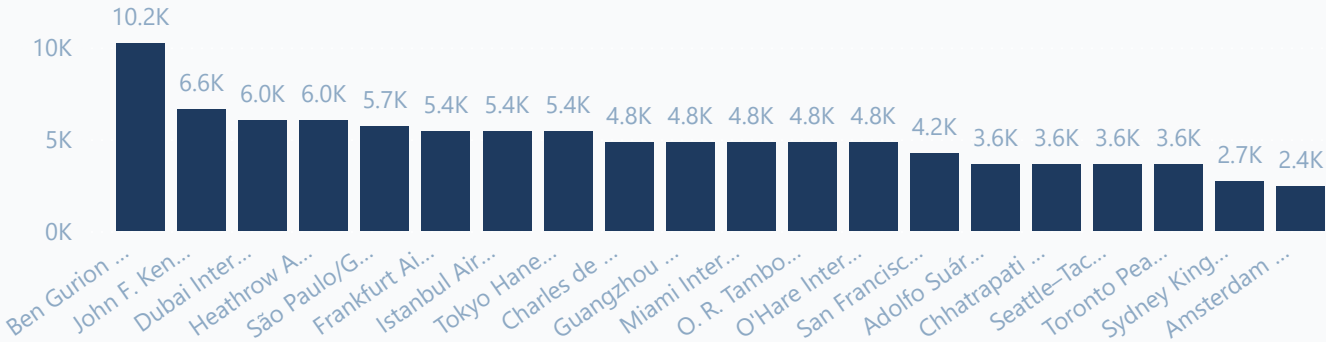
Avg Ticket Price

Revenue Distribution & Analysis

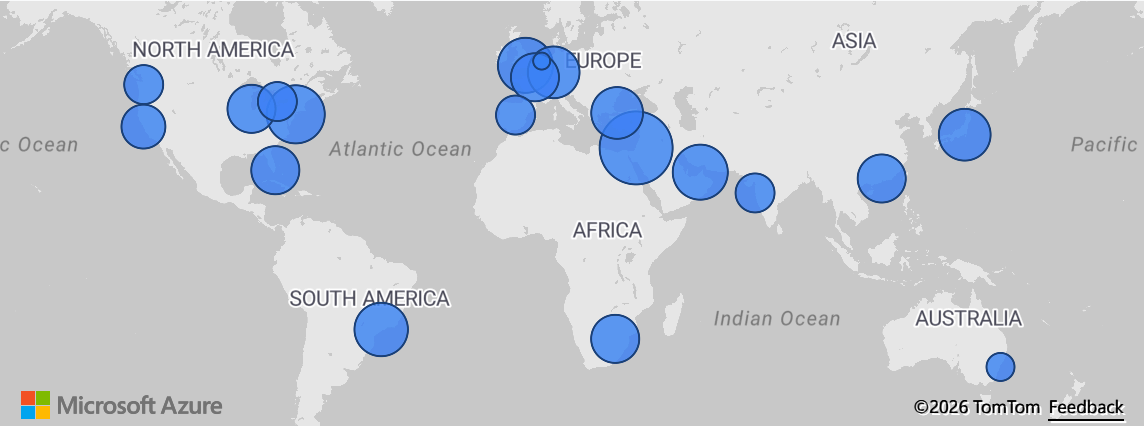
Arrival Airport

Departure Airport

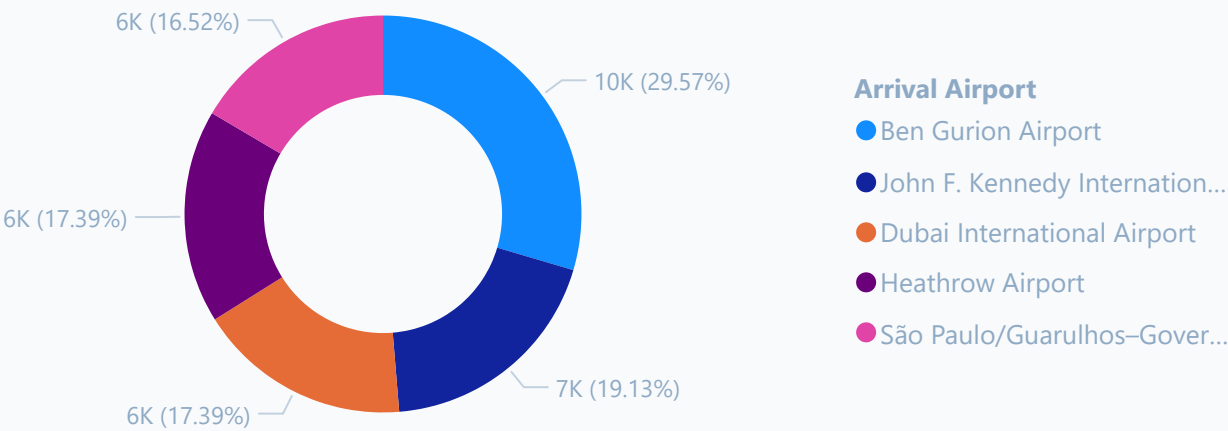
Total Revenue by Arrival Airport



Arrival Airport



Total Revenue by Arrival Airport



Departure Airport	Arrival Airport	Total Revenue
John F. Kennedy International Airport	Ben Gurion Airport	3300
Ben Gurion Airport	Heathrow Airport	2400
San Francisco International Airport	São Paulo/Guarulhos–Governador André Franco Montoro Airport	2400
Adolfo Suárez Madrid–Barajas Airport	Ben Gurion Airport	1800
Ben Gurion Airport	Tokyo Haneda Airport	1800
Dubai International Airport	Toronto Pearson International Airport	1800
Frankfurt Airport	Miami International Airport	1800
Guangzhou Baiyun International Airport	Frankfurt Airport	1800
John F. Kennedy International Airport	Heathrow Airport	1800
Miami International Airport	Charles de Gaulle Airport	1800
San Francisco International Airport	Istanbul Airport	1800

180

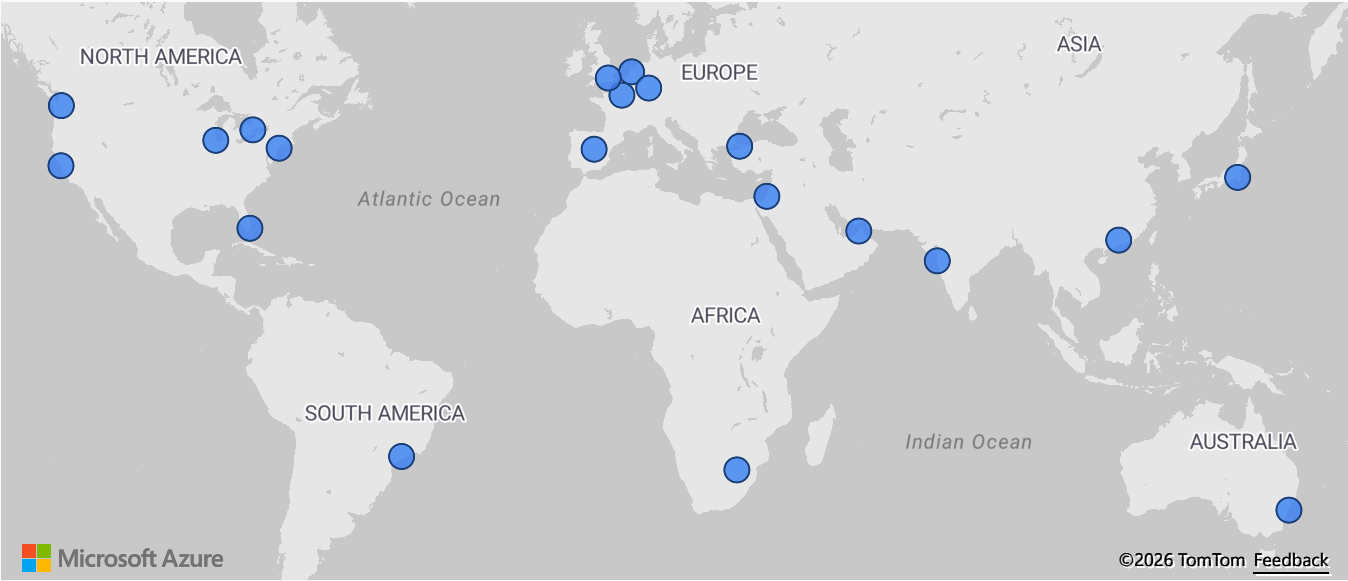
Total Passengers

180

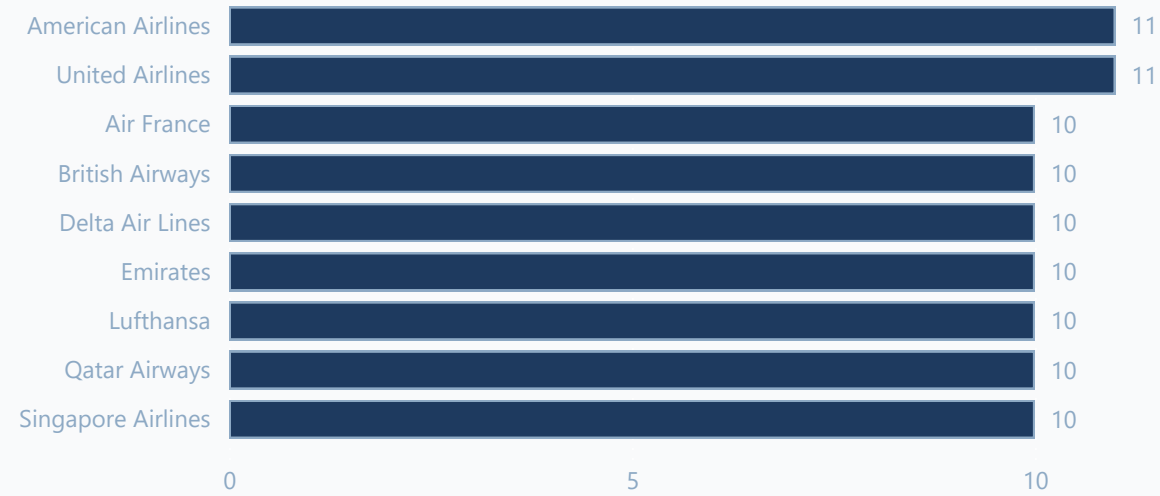
Total Bookings

Passenger Distribution & Travel Patterns

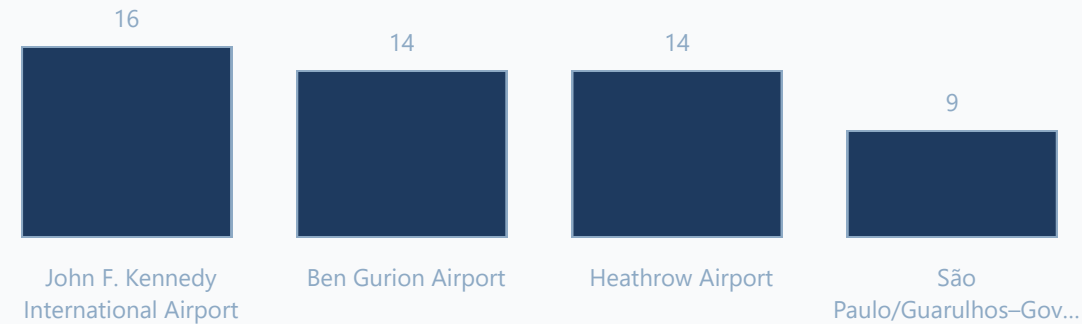
Passenger Locations



Top Airlines With The Most Passengers



Total Passengers by Airport (Top 4)



Top Routes

Departure Airport	Arrival Airport	Total Passengers
Ben Gurion Airport	John F. Kennedy International Airport	5
Ben Gurion Airport	Heathrow Airport	4
Heathrow Airport	John F. Kennedy International Airport	4
John F. Kennedy International Airport	Heathrow Airport	4
Heathrow Airport	Ben Gurion Airport	3
John F. Kennedy International Airport	Ben Gurion Airport	3

180

Total Bookings

Booking Behavior & Trends

